

Debanjan Pal

Senior Game Developer | VR & XR Specialist

4.5+ Years Exp • Unity 2D/3D • Meta Quest VR • C# Architecture • Motion Sensors

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SCAN PORTFOLIO

PROFESSIONAL SUMMARY

Results-driven **Senior Game Developer** with **4.5+ years** of professional industry experience architecting high-performance 2D/3D games, virtual reality medical simulations, and custom hardware motion-control systems. Highly proficient in Unity (URP, Physics, Cinemachine), clean decoupled C# architecture, and Meta Quest 2/3/3S deployment. Recognized for rapid gameplay prototyping, optimizing memory/draw calls for locked high frame rates, and delivering immersive interactive applications from concept to production.

PROFESSIONAL EXPERIENCE

Senior Game Developer — Equidor Medtech | Bengaluru, India

April 2024 – Present

- Lead gameplay programming for cross-platform 2D/3D titles and interactive VR experiences, driving core loops from whitebox to release with sustained 60+ FPS stability.
- Architect specialized applications interfacing directly with proprietary motion-based sensor hardware and custom controllers, cutting input latency across platforms.
- Champion clean engineering workflows, sprint planning, peer code reviews, profiling, and memory optimization across devices, minimizing crashes and draw calls.

Associate Software Developer — Cyclops Medtech | Bengaluru, India

Jan 2022 – March 2024 (2 yrs 3 mos)

- Engineered, optimized, and shipped medical-grade diagnostic VR applications targeting Meta Quest 2 (Android/XR Toolkit), consistently sustaining 72/90 FPS.
- Analyzed telemetry data, diagnostic sensor logs, and system crash dumps to identify technical bottlenecks, accelerate bug resolution, and maintain peak stability.
- Collaborated closely with product owners and clinicians to translate complex clinical protocols into intuitive, gamified patient tests across multiple hardware iterations.

TECHNICAL SKILLS & ARSENAL

GAME DEV & ENGINES

Unity (2D/3D), Universal Render Pipeline (URP), 2D/3D Physics, Cinemachine, UI Toolkit & Canvas, Mecanim Animation, Particle Systems.

VR & HARDWARE SYSTEMS

Meta Quest 2 / 3 / 3S, XR Interaction Toolkit, Motion Sensors, Spatial 6DoF Tracking, Custom Controllers, Medical-Grade VR, Telemetry.

LANGUAGES & ARCHITECTURE

C# (.NET), Design Patterns, State Machines (FSM), Object Pooling, Observer/Events, Data Structures, Decoupled Clean Architecture.

TOOLS & 3D CREATION

Blender 3D, Photoshop, Git / GitLab / GitHub, Unity Profiler, Memory Optimization, Draw Call Batching, JIRA & Agile.

FEATURED PRODUCTION & PORTFOLIO PROJECTS

Animal Swipe — Mobile Gesture & Reflex Arcade

[Demo / Code ↗](#)

Unity

C#

Touch Gestures

Mobile

- Engineered responsive multi-touch swipe mechanics with gesture recognition, combo scoring, and object pooling tailored for locked 60 FPS mobile performance.

Asteroid Run — 3D Space Obstacle Runner

[Google Play ↗](#)

Unity 3D

C#

Procedural Spawning

Android

- Published on Google Play. Built procedural endless asteroid generation with progressive velocity curves, optimized collision matrices, and draw call batching.

Teddy Dispatcher — 3D Physics & Queue Simulation

[Demo / Code ↗](#)

Unity 3D

C#

FSM State Machine

Physics

- Architected an event-driven dispatch loop using State Machines (FSM) to manage dynamic entity routing, sorting queues, and real-time rigid-body interactions.

Walk & Dodge — 3D Obstacle Course Platformer

[Demo / Code ↗](#)

Unity 3D

C#

Physics

Cinemachine

- Engineered 3D obstacle avoidance with dynamic hazards, responsive character physics, Cinemachine tracking, and zero-allocation object pooling locking 60 FPS.

EDUCATION, CREDENTIALS & LANGUAGES

DEGREE

BCA in Game & Mobile App Development

CMR University, Bangalore

2019 – 2022

CERTIFICATION

VR Development Fundamentals in Unity

Udemy Professional Credential

2022

SPOKEN LANGUAGES

English (Professional)

Hindi (Fluent) • Bengali (Native)